6. The hill in the form of a right triangle has its foot at $(19,3)$. The inclination of the hill to the ground is 45°. Find the equation of the hill joining the foot and top.

7. Find the equation of a line through the given pair of points
   (i) $\left(2,\frac{2}{3}\right)$ and $\left(\frac{-1}{2},-2\right)$ (ii) $(2,3)$ and $(-7,-1)$

8. A cat is located at the point $(-6,-4)$ in xy plane. A bottle of milk is kept at $(5,11)$. The cat wish to consume the milk travelling through shortest possible distance. Find the equation of the path it needs to take its milk.

9. If the vertices of a ΔABC are $A(6,2)$, $B(-5,-1)$ and $C(1,9)$; through the vertex A
   (i) find the equation of median (ii) find the equation of altitude

10. Find the equation of a straight line which has slope $\frac{-5}{4}$ and passing through the point $(-1,2)$.

11. You are downloading a song. The percent y (in decimal form) of mega bytes remaining to get downloaded in x seconds is given by $y = -0.1x + 1$.
   (i) find the total MB of the song.
   (ii) after how many seconds will 75% of the song gets downloaded?
   (iii) after how many seconds the song will be downloaded completely?

12. Find the equation of a line whose intercepts on the x and y axes are given below.
   (i) $4, -6$ (ii) $-5, \frac{3}{4}$

13. Find the intercepts made by the following lines on the coordinate axes.
   (i) $3x - 2y - 6 = 0$ (ii) $4x + 3y + 12 = 0$

14. Find the equation of a straight line
   (i) passing through $(1,-4)$ and has intercepts which are in the ratio $2:5$
   (ii) passing through $(-8,4)$ and making equal intercepts on the coordinate axes

5.6 General Form of a Straight Line

The linear equation (first degree polynomial in two variables x and y) $ax + by + c = 0$ (where a, b and c are real numbers such that at least one of a, b is non-zero) always represents a straight line. This is the general form of a straight line.

Now, let us find out the equations of a straight line in the following cases

(i) parallel to $ax + by + c = 0$
(ii) perpendicular to $ax + by + c = 0$

5.6.1 Equation of a line parallel to the line $ax + by + c = 0$

The equation of all lines parallel to the line $ax + by + c = 0$ can be put in the form $ax + by + k = 0$ for different values of k.

5.6.2 Equation of a line perpendicular to the line $ax + by + c = 0$

The equation of all lines perpendicular to the line $ax + by + c = 0$ can be written as $bx - ay + k = 0$ for different values of k.

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